POWER SERIES METHODS

WHY WE NEED SERIES SOLUTIONS OF ODES & SPECIAL FUNCTIONS?



SHORT ANSWER:
Because nice functions ran out!

LINEAR ODES

WITH CONSTANT COEFFICIENTS CAN BE SOLVED ALGEBRAICALLY

EXAMPLE:
$a y'' + b y' + c y = 0$
where a, b, c are constants

WITH VARIABLE COEFFICIENTS

- ARE HARDER
- THEIR SOLUTIONS ARE OFTEN "NON-ELEMENTARY" FUNCTIONS

EXAMPLE:
$a(x) y'' + b(x) y' + c(x) y = 0$
where coefficients $a(x), b(x)$ & $c(x)$ are polynomial functions

KEY EXAMPLES OF SUCH ODEs

APPLICATIONS IN ACOUSTICS, HEAT & FLUID FLOW AND ELECTROMAGNETIC RADIATION

LEGENDRE'S EQUATION

$(1-x^2)y'' - 2xy' + n(n+1)y = 0$
{where n, is a parameter}

BESSEL'S EQUATION

$x^2 y'' + xy' + (x^2 - n^2)y = 0$
{where n, is a parameter}

HYPERGEOMETRIC EQUATION

$x(1-x)y'' + [c-(a+b+1)x]y' - ab\,y = 0$
{where a, b, c are parameters}

SERIES SOLUTION METHOD

Standard method for solving linear ODEs with variable coefficients

TWO STANDARD SOLUTION METHODS

POWER SERIES METHOD

The solution is a power series:

$$y = C_0 + C_1 x + C_2 x^2 + C_3 x^3 + \cdots = \sum_{n=0}^{\infty} C_n x^n$$

FROBENIUS METHOD

- Extends the power series method
- The solution is a power series multiplied by $\ln(x)$ or a fractional power, x^r.
- It's needed when the power series method isn't general enough, such as for Bessel's equation.

Some terminology about summation with an example:

$$\sum_{n=0}^{\infty} f(x)\, x^n$$

- ∞ → going on to a large number
- $n=0$ → starting at $n=0$
- summing or adding each expression
- This expression when expanded looks like this:

$$f(x)(x^0) + f(x)(x^1) + f(x)(x^2) + f(x)(x^3) + \cdots$$

KVB

Power series representation of some elementary functions:

		Converges for
$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$		all x
$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$		all x
$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$		all x
$\cosh(x) = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$		all x
$\sinh(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$		all x
$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$		$-1 < x \leq 1$
$1/(1-x) = 1 + x + x^2 + x^3 + \cdots = \sum_{n=0}^{\infty} x^n$		$-1 < x < 1$



EXAMPLE: Solve $y' - y = 0$

STEP 1 Write y as a power series

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots = \sum_{n=0}^{\infty} c_n x^n$$

STEP 2 Write y' as a term-wise differentiation of y

$$y' = c_1 + 2c_2 x + 3c_3 x^2 + \cdots = \sum_{n=1}^{\infty} n c_n x^{n-1}$$

STEP 3 Substitute into ODE

$$-y = -c_0 - c_1 x - c_2 x^2 - c_3 x^3 - \cdots = -\sum_{n=0}^{\infty} c_n x^n$$

$$y' = c_1 + 2c_2 x + 3c_3 x^2 + 4c_4 x^3 + \cdots = \sum_{n=1}^{\infty} n c_n x^{n-1}$$

$+$

$$y' - y = (c_1 - c_0) + (2c_2 - c_1)x + (3c_3 - c_2)x^2 + \cdots$$

$$\Rightarrow \quad 0 = (c_1 - c_0) + (2c_2 - c_1)x + (3c_3 - c_2)x^2 + \cdots$$

$$\Rightarrow \quad c_1 - c_0 = 0 \qquad \Rightarrow \quad c_1 = c_0$$

$$\Rightarrow \quad 2c_2 - c_1 = 0 \qquad \Rightarrow \quad c_2 = \frac{c_1}{2} = \frac{c_0}{2!}$$

$$\Rightarrow \quad 3c_3 - c_2 = 0 \qquad \Rightarrow \quad c_3 = \frac{c_2}{3} = \frac{c_0}{(3)(2!)} = \frac{c_0}{3!}$$

$\vdots$ $\qquad$ $\vdots$

We have expressions of each coefficient in the power series of y in Step 1

STEP 4

Recall from Step 1:

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots$$

From Step 3:

$$y = c_0 + c_0 x + \frac{c_0}{2!} x^2 + \frac{c_0}{3!} x^3 + \cdots$$

$$= c_0 \left[1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \right]$$

This is the power series representation of e^x

$$\therefore \boxed{y = c_0 e^x}$$

Derive an elementary power series from Taylor and MacLaurin Series!

Def^n:

(#) If any function, $f(x)$ is infinitely differentiable at a point $x=a$, it's Taylor series with center, a is a shown below:

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots$$

Def^n:

Setting, $a=0$ gives the MacLaurin series as shown below;

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

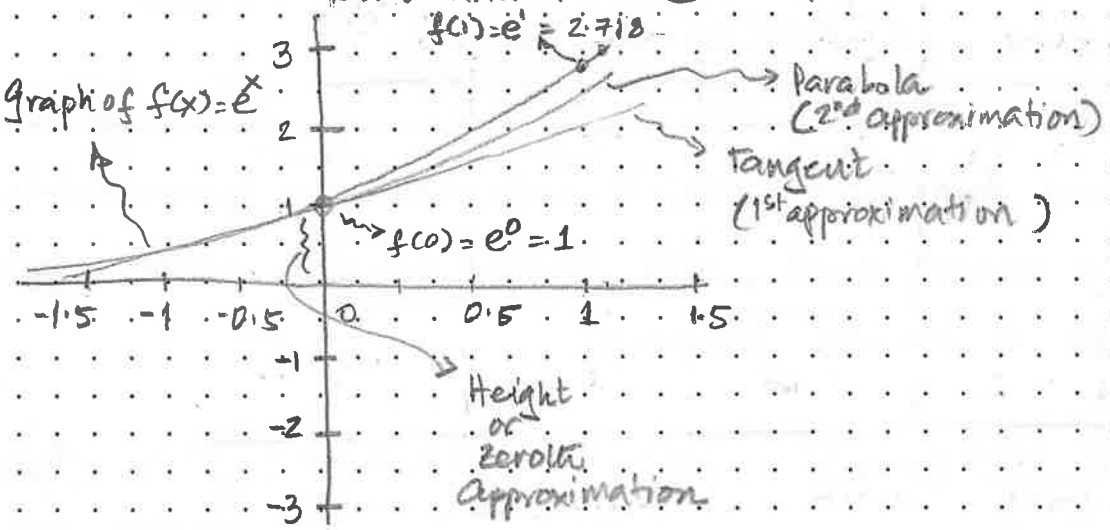
(#) The coefficients of the Power series representation are fixed: $\boxed{c_n = \frac{f^{(n)}(0)}{n!}}$

(#) In general, we are asking the question: If we know what's happening with our function at some point x, i.e., $f(x)$ can we find out what's happening at $(x+\Delta x)$, i.e., $f(x+\Delta x)$?

$$\Rightarrow f(x+\Delta x) \approx \underbrace{f(x)}_{\text{Height}} + \underbrace{(\Delta x)\frac{df(x)}{dx}}_{\substack{1^{st}\text{ derivative.} \\ \text{or slope or tangent} \\ 1^{st}\text{ CORRECTION TERM}}} + \underbrace{\frac{1}{2}(\Delta x)^2\frac{d^2f(x)}{dx^2}}_{\substack{2^{nd}\text{ derivative,} \\ \text{bends the tangent} \\ 2^{nd}\text{ CORRECTION TERM}}} + \cdots$$

EXAMPLE: TAYLOR SERIES EXPANSION OF $f(x) = e^x$

with known value at $x = 0$



	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
The function & its derivatives	e^x	e^x	e^x	e^x
At $x = 0$	$e^0 = 1$	$e^0 = 1$	$e^0 = 1$	$e^0 = 1$

Taylor series at $x = a \Rightarrow$ MacLaurin series

$$\therefore e^x = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

$$= 1 + (1)(x) + \left(\frac{1}{2!}\right)x^2 + \left(\frac{1}{3!}\right)x^3 + \cdots$$

$$\sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$$

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots$$

Power series of e^x

Height or Zeroth approximation of e^x

Slope with 1st approximation of e^x

Slope of a slope OR Best slope or Parabola; 2nd approximation

Cubic approximation of e^x

Together it's a line or tangent; Approximation of e^x

Parabolic approximation of e^x

EXAMPLE: Derive the power series of $f(x) = \cos(x)$

	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
The function & it's derivatives	$\cos(x)$	$-\sin(x)$	$-\cos(x)$	$\sin(x)$
$x=0$	$\cos(0) = 1$	$-\sin(0) = 0$	$-\cos(0) = -1$	$\sin(0) = 0$

Taylor series at $x=0$ $\Rightarrow$ Maclaurin series

$$\cos(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

$$\Rightarrow \cos(x) = 1 + (0)(x) + \frac{(-1)(x^2)}{2!} + \frac{(0)(x^3)}{3!} + \cdots$$

$$\Rightarrow \boxed{\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots} \Leftarrow \text{Power series of } \cos(x)$$

In summation notation:

$$\boxed{\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}}$$

THE SEVEN STEP RECIPE

STEP-1 Differentiate $f(x)$ repeatedly:

$f, f', f'', f''', \ldots$

STEP-2 Evaluate each derivative at $x = 0$:

$f(0), f'(0), f''(0), f'''(0), \ldots$

STEP-3 Substitute in the Maclaurin series for $f(x)$:

$$f(0) + f'(0)(x) + \frac{f''(0)}{2!}(x^2) + \frac{f'''(0)}{3!}(x^3) + \ldots$$

STEP-4 Write a general formula for $f^{(n)}(0)$:

STEP-5 Find the formula for the coefficients:

$$c_n = \frac{f^{(n)}(0)}{n!}$$

Recurrence relation

STEP-6 Write down the power series in its summation form:

$$\sum_{n=0}^{\infty} c_n x^n = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

STEP-7 Evaluate convergence of the power series.

(i) Does the series converge?

(ii) Does the series converge to f?

THE POWER SERIES METHOD

→ The power series method of solving a differential equation consists of

(i) substituting the power series

$$y = \sum_{n=0}^{\infty} c_n x^n$$

in the differential equation, and then

(ii) determing the coefficients, $c_0, c_1, c_2, \ldots$ that satisfy the differential equation.

→ Depending on the order of the differential equation, we substitute for y and its derivatives, $y', y'', y''', \ldots$ i.e.,

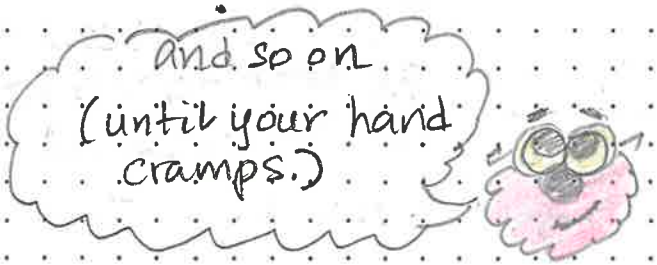
$$y = \sum_{n=0}^{\infty} c_n x^n$$

$$y' = \sum_{n=1}^{\infty} n c_n x^{n-1} = \sum_{n=0}^{\infty} (n+1) c_{n+1} x^n$$

$$y'' = \sum_{n=2}^{\infty} (n)(n-1) c_n x^{n-2} = \sum_{n=0}^{\infty} (n+2)(n+1) c_{n+2} x^n$$

⋮

"shifting the index"
These summations are adjusted or "shifted" to start at $n = 0$



Convergence of Power Series

① When does a power series exist?

Answer: (1) When $A(x), B(x), C(x), D(x)$ in the ODE

$$A(x)y'' + B(x)y' + C(x)y = D(x)$$

have power series representations (i.e. Taylor series),

(2) More precisely, the functions $A(x), B(x), C(x), D(x)$ are said to be

analytic at some point $x = a$ if their Power (Taylor) series,

converge for all x in some interval containing the point a

② If two functions, (in general) $f(x)$ & $g(x)$ are both analytic at $x = a$, then so are their

(i) sum: $f(x) + g(x)$ — In terms of power series: Termwise addition

(ii) Product: $f(x) \cdot g(x)$ — Formal multiplication

(iii) quotient: $\frac{f(x)}{g(x)}$ if $g(x) \neq 0$ — Formal / Long Division

(iv) derivative: $f'(x)$ — Termwise differentiation

EXAMPLE OF FORMAL MULTIPLICATION

$$\underbrace{\sin(x)}_{f(x)}\ \underbrace{\cos(x)}_{g(x)} = \left(x - \frac{1}{6}x^3 + \frac{1}{120}x^5 - \cdots\right)\left(1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 - \cdots\right)$$

$$= x - \frac{1}{6}x^3 + \frac{1}{120}x^5 - \cdots$$
$$- \frac{1}{2}x^3 + \frac{1}{12}x^5 - \frac{1}{240}x^7 + \cdots$$
$$+ \frac{1}{24}x^5 - \cdots$$
$$= x + \left(-\frac{1}{6} - \frac{1}{2}\right)x^3 + \left(\frac{1}{120} + \frac{1}{12} + \frac{1}{24}\right)x^5 + \cdots$$
$$= x - \frac{4}{6}x^3 + \frac{16}{120}x^5 - \cdots$$
$$= \frac{1}{2}\left[(2x) - \frac{(2x)^3}{6} + \frac{(2x)^5}{120} - \cdots\right]$$
$$= \frac{1}{2}\left[(2x) - \frac{(2x)^3}{3!} + \frac{(2x)^5}{5!} - \cdots\right]$$
$$= \frac{1}{2}\sin(2x) \quad \checkmark$$

EXAMPLE OF DIVISION OF POWER SERIES TO GET tan(x)

$$\tan(x) = \frac{\sin(x)}{\cos(x)} = \frac{x - x^3/6 + x^5/120 - x^7/5040 + \cdots}{1 - x^2/2 + x^4/24 - x^6/720 + \cdots}$$

$$x + x^3/3 + 2x^5/15 + \cdots$$
$$1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \cdots \;\Big)\; x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} + \cdots$$
$$- \left(x - \frac{x^3}{2} + \frac{x^5}{24} - \frac{x^7}{720} + \cdots\right)$$
$$\frac{x^3}{3} - \frac{x^5}{30} + \frac{x^7}{840} + \cdots$$
$$- \left(\frac{x^3}{3} - \frac{x^5}{6} + \frac{x^7}{72} + \cdots\right)$$
$$\frac{2x^5}{15} - \frac{4x^7}{315} + \cdots$$

(3) How can I be sure of convergence of a power series? Answer:

One method to test convergence is the ratio test.

For a given power series,

$$y = \sum_{n=0}^{\infty} c_n x^n$$

suppose the limit,

$$\boxed{R = \lim_{n\to\infty} \left| \frac{c_n}{c_{n+1}} \right|}$$

exists,

exists (R is finite) or is infinite ($R = \infty$), then

(i) If $\boxed{R = 0}$, the series diverges for all $x \neq 0$

(ii) If $\boxed{0 < R < \infty}$, the series converges if $|x| < R$

diverges if $|x| > R$

(iii) If $\boxed{R = \infty}$, the series converges for all x.

EXAMPLE: Derive the power series of $f(x)=\sin(x)$ and confirm its convergence.

STEPS 1 & 2

	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$	$f^{(iv)}(x)$
The function & its derivatives	$\sin(x)$	$\cos(x)$	$-\sin(x)$	$-\cos(x)$	$\sin(x)$
At $x=0$	$\sin(0)=0$	$\cos(0)=1$	$-\sin(0)=0$	$-\cos(0)=-1$	$\sin(0)=0$

STEP-3 Taylor series at $x=0$ $\Rightarrow$ MacLaurin series

$$\sin(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

$$= 0 + (1)(x) + \frac{(0)x^2}{2!} + \frac{(-1)}{3!}x^3 + \left(\frac{0}{4!}\right)x^4 + \frac{(1)x^5}{5!} + \cdots$$

$$\therefore \boxed{\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots} \Leftarrow \text{Power series.}$$

STEP-4 General formula for, $f^{(n)}(0)$:

Pattern for $f^{(n)}(0)$: $0, 1, 0, -1, 0, 1 \Rightarrow$ Only odd powers of x have coefficients!

$$\boxed{f^{(2n+1)}(0) = (-1)^n}$$

STEP-5 Formula for coefficients, c_n: [Recurrence relation]

$$c_n = \frac{f^{(n)}(0)}{n!} \Rightarrow c_n = \frac{f^{(2n+1)}(0)}{(2n+1)!} \Rightarrow \boxed{c_n = \frac{(-1)^n}{(2n+1)!}}$$

STEP-6 Power series as a summation!

$$\sum_{n=0}^{\infty} c_n x^n = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n \Rightarrow \boxed{\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}}$$

$$= \sum_{n=0}^{\infty} \frac{f^{(2n+1)}(0)}{(2n+1)!}x^{2n+1}$$

STEP 7 CONVERGENCE TEST:

$$\lim_{n\to\infty}\left|\frac{c_n}{c_{n+1}}\right| = R$$ ⇝ radius of convergence

⇝ $R = \infty, 1, 0$

⇝ If the limit exists and, $R=\infty$, then the power series converges for all values of x, in the power series [Best possible case]

⇝ If the limit exists and, $R=1$, then the power series converges in some finite interval of x, in the power series [Usual case]

⇝ If the limit exists and, $R=0$, then the power series converges only at the center, or the point where the series is evaluated. [Useless case]

[Convergence of power series of $\sin(x)$]

$c_n = \dfrac{(-1)^n}{(2n+1)!}$ from step-5

$c_{n+1} = \dfrac{(-1)^{n+1}}{(2(n+1)+1)!} = \dfrac{(-1)^{n+1}}{(2n+3)!}$

$$\left|\frac{c_n}{c_{n+1}}\right| = \left|\frac{(-1)^n}{(2n+1)!}\,\frac{(2n+3)!}{(-1)^{n+1}}\right| = \left|\frac{(2n+3)(2n+2)\cancel{(2n+1)!}}{\cancel{(2n+1)!}}\right|$$

$$\lim_{n\to\infty}\left|\frac{c_n}{c_{n+1}}\right| = \lim_{n\to\infty}(2n+3)(2n+2) = \infty$$

$\Rightarrow \boxed{R=\infty} \Rightarrow \sin(x)$ converges for all x.

EXAMPLE: Solve the equation $y'' + y = 0$ using power series representation for y.

Solution:

STEP-1 Write the power series representation of y:

$$y = \sum_{n=0}^{\infty} c_n x^n$$

STEP-2 Termwise differentiation:

$$y' = \sum_{n=1}^{\infty} n c_n x^{n-1} \Rightarrow y'' = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}$$

By shifting the index $n=2$ to $n=0$

$$y'' = \sum_{n=0}^{\infty} (n+2)(n+1) c_{n+2} x^n$$

STEP-3 Substitute into $y'' + y = 0$

$$\sum_{n=0}^{\infty} (n+2)(n+1) c_{n+2} x^n + \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} \left[(n+2)(n+1) c_{n+2} + c_n \right] x^n = 0$$

For the above equation to hold for all x, all coefficients must vanish / disappear or go to zero.

$$\Rightarrow (n+2)(n+1) c_{n+2} + c_n = 0$$

STEP-4 Write the recurrance relation:

$$c_{n+2} = -\frac{c_n}{(n+2)(n+1)} \qquad n \geq 0$$

STEP-5 Compute the coefficients:

Even terms

$n = 0, 2, 4, \ldots$

$n=0$: $c_2 = -\frac{c_0}{2} = \frac{-c_0}{2!}$

$n=2$: $c_4 = \frac{-c_2}{(4)(3)} = \frac{-c_0}{4!}$

$n=4$: $c_6 = \frac{-c_4}{(6)(5)} = \frac{-c_0}{6!}$

[where c_0 is an arbitrary constant]

$\Rightarrow c_{2k} = \frac{(-1)^k c_0}{(2k)!}$

Odd terms

$n = 1, 3, 5, \ldots$

$n=1$: $c_3 = -\frac{c_1}{(3)(2)} = \frac{-c_1}{3!}$

$n=3$: $c_5 = \frac{-c_3}{(5)(4)} = \frac{-c_1}{5!}$

$n=5$: $c_7 = \frac{-c_5}{(7)(6)} = \frac{-c_1}{7!}$

[where c_1 is an arbitrary constant]

$\Rightarrow c_{2k+1} = \frac{(-1)^k c_1}{(2k+1)!}$

STEP-6 Assemble the solution:

$$y(x) = \sum_{k=0}^{\infty} c_{2k} x^{2k} + \sum_{k=0}^{\infty} c_{2k+1} x^{2k+1}$$

Even terms; Odd terms

$$= c_0 \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k} + c_1 \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1}$$

$$= c_0 \left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\right] + c_1 \left[x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right]$$

Maclaurin series for cosine; Maclaurin series for sine

$$= c_0 \cos(x) + c_1 \sin(x)$$

$$\therefore \boxed{y(x) = c_0 \cos(x) + c_1 \sin(x)}$$

STEP-7 Check for convergence:

(#b) For the series to converge for all x, both even and odd parts should converge!

Ratio test on even terms

$$R_1 = \lim_{m\to\infty} \left|\frac{c_m}{c_{m+1}}\right|$$

$$= \lim_{m\to\infty} \left|\frac{(-1)^m/(2m)!}{(-1)^{m+1}/[2(m+1)]!}\right|$$

$$= \lim_{m\to\infty} \left|-\frac{(2m+2)!}{(2m)!}\right|$$

$$= \lim_{m\to\infty} (2m+2)(2m+1)\frac{(2m)!}{(2m)!}$$

$$= \infty \checkmark$$

Ratio test on odd terms

$$R_2 = \lim_{p\to\infty} \left|\frac{c_p}{c_{p+1}}\right|$$

$$= \lim_{p\to\infty} \left|\frac{(-1)^p/(2p+1)!}{(-1)^{p+1}/[2(p+1)+1]!}\right|$$

$$= \lim_{p\to\infty} \left|-\frac{(2p+3)!}{(2p+1)!}\right|$$

$$= \lim_{p\to\infty} (2p+3)(2p+2)$$

$$= \infty \checkmark$$

$R_1 = R_2 = \infty$

$\Rightarrow$ The solution converges for all x